

 GUIDE		PLG-GD-0016-V1	
Low Short Circuit Strength Screening for Inverter-based Resources		Version:	1
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1. Purpose

The purpose of this methodology is to define the process by which ATC screens the system for low short circuit strength conditions that will require an Electromagnetic Transients (EMT) study.

2. Scope

This methodology only applies to the ATC system and inverter-based resources that connect to the ATC system.

3. Applicability

3.1 Regulatory Applicability

NERC Reliability Standards FAC-002-2, TPL-001-4

3.2 Affected Assets

Inverter-based resources that are connected to the ATC system and the transmission facilities that may be affected by the interconnection

3.3 Affected Personnel

ATC System Planning staff

4. Methodology

4.1 Short Circuit Ratio Calculations

ATC uses two calculation methods to assess the short circuit strength of the grid where inverter-based resources are interconnected, Short Circuit Ratio (SCR) and Weighted Short Circuit Ratio (WSCR).

SCR only takes into account one inverter-based resource and is calculated as follows:

$$SCR_{POI} = \frac{SCMVA_{POI}}{MW}$$

Where,

SCR_{POI} is the SCR calculated at the Point of Interconnection (POI)

$SCMVA_{POI}$ is the short circuit MVA magnitude calculated at the POI

MW is the full MW capability of the inverter-based generating facility

WSCR takes into account multiple inverter-based resources in close proximity and is calculated as follows:

$$WSCR = \frac{\sum_i^N SCMVA_i * P_{RMWi}}{(\sum_i^N P_{MWi})^2}$$

Where,

WSCR is the weighted SCR of a group of inverter-based resources

SCMVA_i is the short circuit MVA magnitude calculated at the POI of the
ith inverter-based generating facility in the group

P_{MWi} is the full MW capability of the ith inverter-based generating facility

When taking outages to determine the WSCR of a group of generators the
outage cannot isolate the generators from each other.

4.2 When to Screen for Low Short Circuit Strength

ATC is concerned with control instability resulting from the interconnection of inverter-based resources in areas of the grid with low short circuit strength. Given this concern a low short circuit strength screening study should be performed when any of the following changes are proposed on the transmission system:

1. Interconnection of a new inverter-based resource.
2. Retirement of a large synchronous machine that is near existing inverter-based resources or near proposed inverter-based resources with signed GIAs.
3. Addition of series capacitance near existing inverter-based resources or near proposed inverter-based resources with signed GIAs.
4. Extended transmission facility outages, removal of significant transmission facilities, or cancellation of significant transmission projects near existing inverter-based resources or near proposed inverter-based resources with signed GIAs.

4.3 Case and Contingency Selection

Low short circuit strength screening will be performed on a PSSE load flow case representing a shoulder load condition. The study year of the case should reflect the time period when the new interconnection or other system change is proposed to be completed.

SCRs will be calculated for system intact and contingency conditions.

Contingencies used in the weak grid screening will include all contingencies that would normally be selected for a stability study per the latest ATC TPL contingency rationale document.

Recommended parameters to use with the PSSE APIs for calculating short circuit MVA are provided in Appendix A.

4.4 Grouping for Weighted Short Circuit Ratio

Determining which generators will be included in a group for calculating WSCR will be determined by engineering judgement. Factors to be considered when making that engineering judgement include:

1. Results from previous studies.
2. Transmission line miles between inverter-based resource POIs.
3. Transmission impedance between inverter-based resource POIs.
4. MW capabilities of nearby inverter-based resource POIs.
5. Inverter-based resources that share the same outlet path back to a strong source.
6. Inverter-based resources on the same kV level path without many connections to strong sources.

4.5 Criteria to Determine if an EMT Study is Needed

An EMT study should be performed if it is determined that an inverter-based resource is interconnecting to a part of the grid with low short circuit strength as defined by any of the following criteria:

1. The SCR is less than manufacturer provided minimum SCR at the POI required for stable operation. If a minimum SCR for stable operation at the POI is not provided a minimum SCR at the POI of 3 for stable operation will be assumed.
2. The WSCR is less than 1.5.
3. The inverter-based resource is interconnecting to the transmission system of Michigan's Upper Peninsula east of ATC's Arnold 345-138 kV Substation due to concerns with potential interactions with the Mackinac HVDC facility.
4. The system in the area of the POI has been found to have low short circuit strength in actual operation or per engineering judgement.

4.6 Faults to Simulate in EMT Study

Given the complexity and cost of an EMT study it is preferred to only run the worst few faults (including both Non-P6 and P6) for each inverter-based resource. The SCRs and WSCRs from the screening analysis should be used to determine the most severe faults to be run in the EMT Study. It is reasonable to

assume that the faults resulting in the lowest SCRs and WSCRs will be the most conservative faults to simulate.

5. Administration

5.1 Review

Review at least every five years.

5.2 Retention

The previous version of this document will be retained for at least 5 years after it is retired.

6. Approvals/Revision History

Revision	Author(s)	Reviewer(s)/ Approver(s)	Effective Date	Summary of Changes
1	<i>Chengyue Guo, Josh Kerr, Mike Marz, P.E., Damien Sommer, P.E.</i>	Paul Walter, P.E.	9/10/2019	Initial document

Appendix A: Recommended PSSE API Parameters for Calculating Short Circuit MVA

Before running either the ASCC_3 or ASCC_CURRENTS Application Program Interfaces (APIs) for PSSE the short circuit units and coordinates should be set as follows:

`psspy.short_circuit_units(1)` # Set short circuit units to physical units

`psspy.short_circuit_coordinates(1)` # Set short circuit coordinates to polar coordinates

Table A.1: ATC Recommended Parameter Settings for PSSE API ASCC_3

Parameter	Description	Options	Recommended Setting
STATUS(1)	line-to-ground faults	0 - omit 1 - include	0
STATUS(2)	line-out faults	0 - omit 1 - include	1
STATUS(3)	line-end faults	0 - omit 1 - include	0
STATUS(4)	bus voltage option	0 - use bus voltages from power flow solution 1 - set all bus voltages at specified value and at 0 deg 2 - set faulted bus voltage at specified value and at 0 deg	1
STATUS(5)	fault analysis generator reactance option; ignored if generators are already converted	0 - use subtransient reactance 1 - use transient reactance 2 - use synchronous reactance	0
STATUS(6)	report option	0 - fault current summary table 1 - total fault currents 2 - fault contributions to N levels away 3 - total fault currents and fault contributions to N levels away	1

STATUS(7)	number of levels back for contributions (used if STATUS(6) = 2 or 3)	specify integer	0
STATUS(8)	transformer tap ratios and phase shift angles option	0 - leave tap ratios and phase shift angles unchanged 1 - set tap ratios to 1.0 pu and phase shift angles to 0 2 - set tap ratios to 1.0 pu and phase shift angles unchanged 3 - set tap ratios unchanged and phase shift angles to 0	1
STATUS(9)	dc line and FACTS device option	0 - blocked 1 - represent as load	0
STATUS(10)	zero sequence transformer impedance correction option	0 - ignore 1 - apply	0
STATUS(11)	three phase faults	0 - omit 1 - include	1
STATUS(12)	line-line-to-ground faults	0 - omit 1 - include	0
STATUS(13)	line-to-line faults	0 - omit 1 - include	0
STATUS(14)	line charging option	0 - leave unchanged 1 - set to 0.0 in the positive and negative sequences 2 - set to 0.0 in all sequences	2
STATUS(15)	line, fixed and switched shunts, and magnetizing admittance option	0 - leave unchanged 1 - set to 0.0 in the positive and negative sequences 2 - set to 0.0 in all sequences	2
STATUS(16)	load option	0 - leave unchanged 1 - set to 0.0 in the positive and negative sequences 2 - set to 0.0 in all sequences	2
STATUS(17)	synchronous and asynchronous machines power option	0 - use real and reactive power outputs from power flow solution 1 - set real and reactive power outputs to 0.0	1
VALUES	User specified pu bus voltage value (Used when STATUS (4) = 1 or 2)	specify value	1

Table A.2: ATC Recommended Parameter Settings for PSSE API ASCC CURRENTS

Parameter	Description	Options	Recommended Setting
flt3ph	Is to report three phase faults	0 - omit 1 - include	1
fltlg	Is to report line-to-ground faults	0 - omit 1 - include	0
fltllg	Is to report line-line-to-ground faults	0 - omit 1 - include	0
ftlll	Is to report line-to-line faults	0 - omit 1 - include	0
linout	Is to report line-out faults	0 - omit 1 - include	1
linend	Is to report line-end faults	0 - omit 1 - include	0
voltop	Is the bus voltage option	0 - use bus voltages from power flow solution 1 - set all bus voltages at specified value and at 0 deg 2 - set faulted bus voltage at specified value and at 0 deg	1
genxop	Is the synchronous machine reactance option, ignored if generators are already “converted”	0 - use subtransient reactance 1 - use transient reactance 2 - use synchronous reactance	0
tpunty	Is the transformer tap ratios and phase angle option	0 - leave tap ratios and phase shift angles unchanged 1 - set tap ratios to 1.0 pu and phase shift angles to 0 2 - set tap ratios to 1.0 pu and phase shift angles unchanged 3 - set tap ratios unchanged and phase shift angles to 0	1
dcload	Is the dc line and FACTS device option	0 - blocked 1 - represent as load	0

zcorec	Is the zero sequence transformer impedance correction option	0 - ignore 1 - apply	0
lnchrg	Is the line charging option	0 - leave unchanged 1 - set to 0.0 in the positive and negative sequences 2 - set to 0.0 in all sequences	2
shntop	Is the line shunts, fixed shunts, switched shunts and transformer magnetizing admittance option	0 - leave unchanged 1 - set to 0.0 in the positive and negative sequences 2 - set to 0.0 in all sequences	2
loadop	Is the load option	0 - leave unchanged 1 - set to 0.0 in the positive and negative sequences 2 - set to 0.0 in all sequences	2
machpq	Is the synchronous and asynchronous machines power output option	0 - use real and reactive power outputs from power flow solution 1 - set real and reactive power outputs to 0.0	1
volts	Is the user specified bus voltage value in pu	specify value	1

